IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

Introduction

Rocket launch costs:

- SpaceX Falcon 9:

  62 million dollars

- other providers:

  upward of 165 million dollars

Reason of difference:

- SpaceX can reuse the first stage.

Purpose of research:

- Determine if first stage will land

Why is it important?:

- Predict the cost of launch
- Give opportunity to other companies to compete with SpaceX

Section 1

Methodology

Methodology

Executive Summary:

- Built a complete Falcon 9 landing-success analysis pipeline.
- Collected launch data from SpaceX REST API and Wikipedia scraping.
- Cleaned landing outcomes into a binary target: Success / Failure.
- Explored patterns with visualization, SQL, and Folium maps.
- Trained classification models and compared accuracy/confusion matrices.

Data Collection

- Data were collected from following sources:
  - api.spacexdata.com
  - https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches
- Key tools/libraries used for data collection:
  - Web scraping:
    - Beautifulsoup4
    - Requests
  - SpaceX REST API:
    - Requests

Data Collection – SpaceX API

- **Data collection with SpaceX REST:**

```
spacex_url = https://api.spacexdata.com/v4/launches/past

response = requests.get(spacex_url)

data = pd.json_normalize(response.json())
```

Data Collection - Scraping

- **Web scraping process using key phrases and flowcharts**

```
response = requests.get(static_url, headers = headers)
soup = BeautifulSoup(response.text)
html_tables = []
for table in soup.find_all('table'):
    html_tables.append(table)
```

External reference:

https://github.com/Neuvox512/Data-Science-Capstone-Project

Data Wrangling

- To determine if landing of first stage was successful ‘Outcome’ column was processed
- From different types of landing were extracted only values 1 and 0, indicating Success and Failure respectively
- These values formed new column and served as target values

External reference:

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EDA with Data Visualization

Charts used in Data Visualization Analysis:

- Relationship between Flight Number and Launch Site (scatter plot)
- Relationship between Payload Mass and Launch Site (scatter plot)
- Relationship between success rate of each orbit type (bar plot)
- Relationship between FlightNumber and Orbit type (scatter plot)
- Relationship between Payload Mass and Orbit type (scatter plot)
- Launch success yearly trend (line chart)

External reference:

https://github.com/Neuvox512/Data-Science-Capstone-Project

EDA with SQL

- SPACEXTABLE was created in sqlite database
- Unique names of launch site were displayed
- Total Payload mass carried by boosters launched by NASA (CRS) was found
- average payload mass carried by booster version F9 v1.1 was found
- date when the first successful landing outcome in ground pad was achieved has been found
- Names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000 were explored
- total number of successful and failure mission outcomes was found
- All the booster versions that have carried the maximum payload mass were explored
- All records which display the month, failure landing in drone ship ,booster versions and launch site for the months in year 2015 were extracted
- All landing cases between the date 2010-06-04 and 2017-03-20 were extracted in descending order.

External reference:

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Build an Interactive Map with Folium

- First folium.Map object was created with coordinates related to NASA Johnson Space Center at Houston, Texas
- Then folium.Circle and folium.Marker object were used to highlight and label NASA Space Center
- Then all Launch Site and their coordinates were extracted from data set
- Launch sites were highlighted the same way with folium.Circle
- MarkerCluster object was used to present successful and unsuccessful landing of rocket's first stage launched from respective launch site
- Then PolyLine object was created to represent the distance between launch site and their proximities

External reference:

https://github.com/Neuvox512/Data-Science-Capstone-Project

Predictive Analysis (Classification)

Model workflow

- Split the dataset into features (X) and target (Y).
- Standardized numeric features with StandardScaler.
- Used an 80:20 train/test split.
- Tuned Logistic Regression, SVM, Decision Tree, and KNN with GridSearchCV.
- Compared accuracy and confusion matrices to select the best model.

- External reference:
- https://github.com/Neuvox512/Data-Science-Capstone-Project

External reference:

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Results

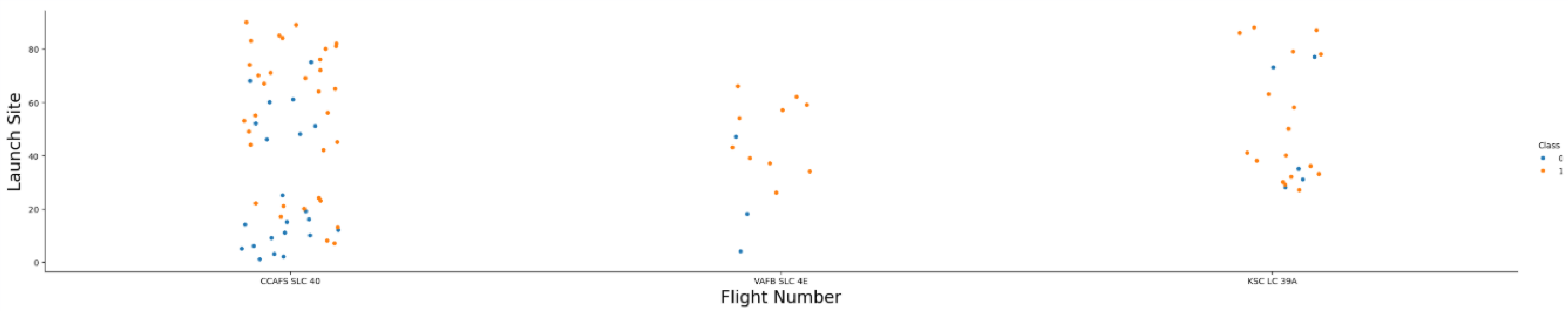
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

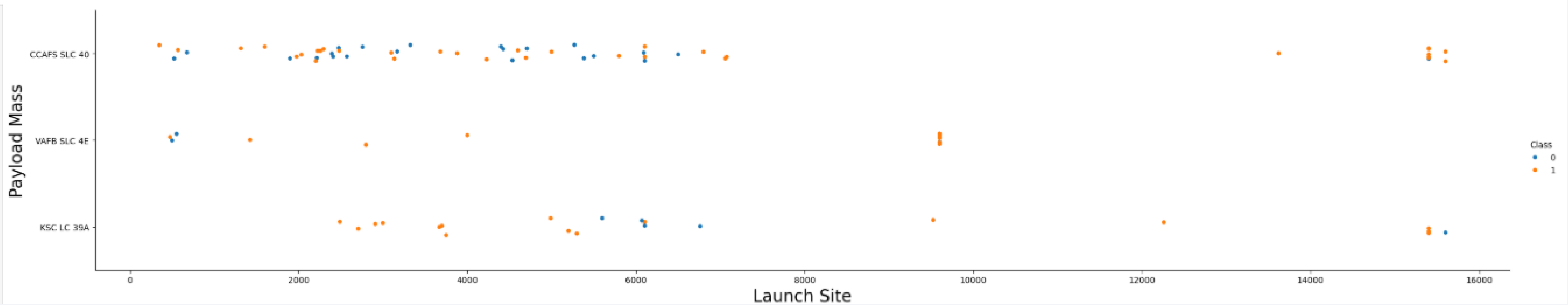
- The chart is now taken from the uploaded notebook output.
- Successful first-stage landings become more common at higher flight numbers.
- CCAFS SLC-40 has the highest number of launches in this visual.



External reference: https://github.com/Neuvox512/Data-Science-Capstone-Project

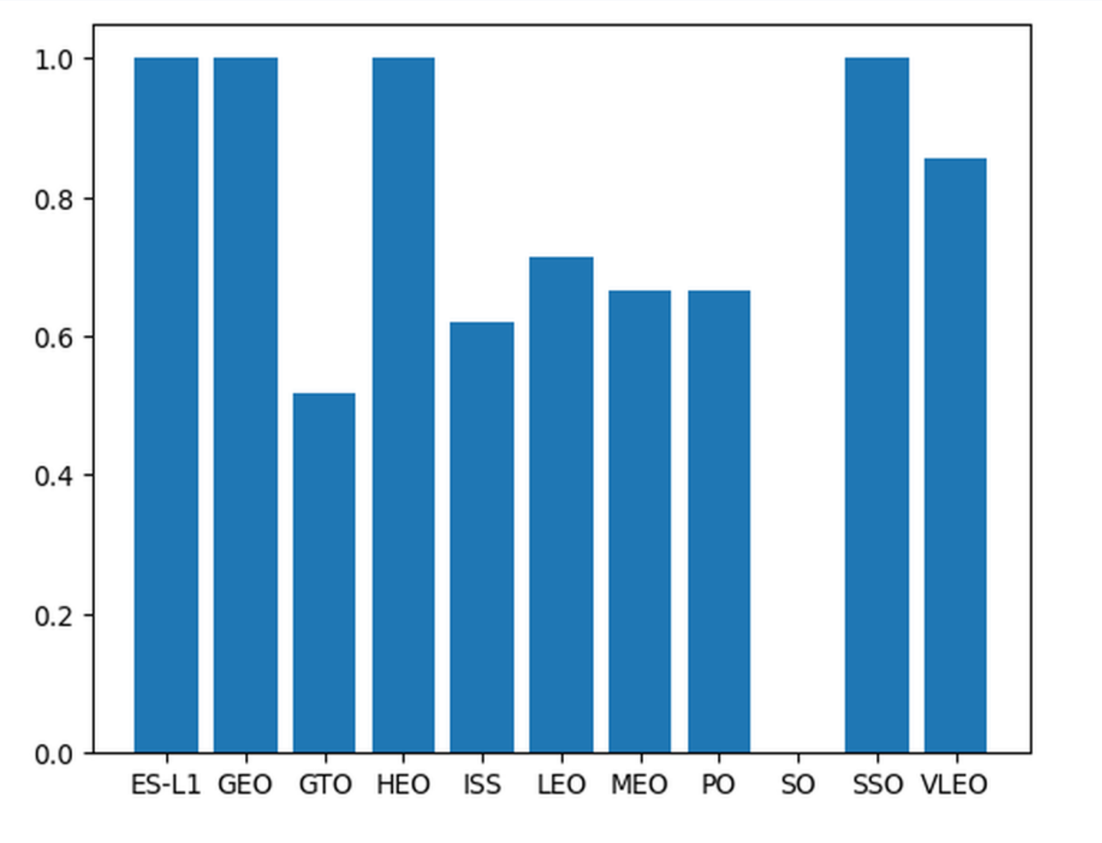
Payload Mass vs. Launch Site

- The chart is now taken from the uploaded notebook output.
- Payload distributions vary by launch site.
- The notebook notes that VAFB SLC-4E does not include very heavy payloads above 10,000 kg.



External reference: https://github.com/Neuvox512/Data-Science-Capstone-Project

Success Rate by Orbit Type

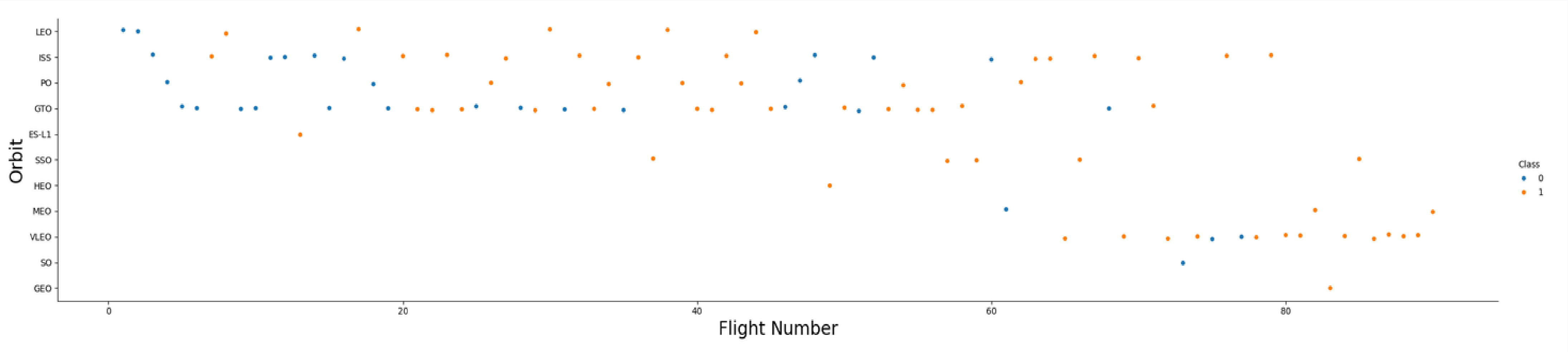


- Orbit type has a strong relationship with landing success.
- GTO shows a visibly lower success rate than several other orbit classes.
- ES-L1, GEO, HEO, and SSO appear near 100% successful in this sample.

External reference: https://github.com/Neuvox512/Data-Science-Capstone-Project

Flight Number vs. Orbit Type

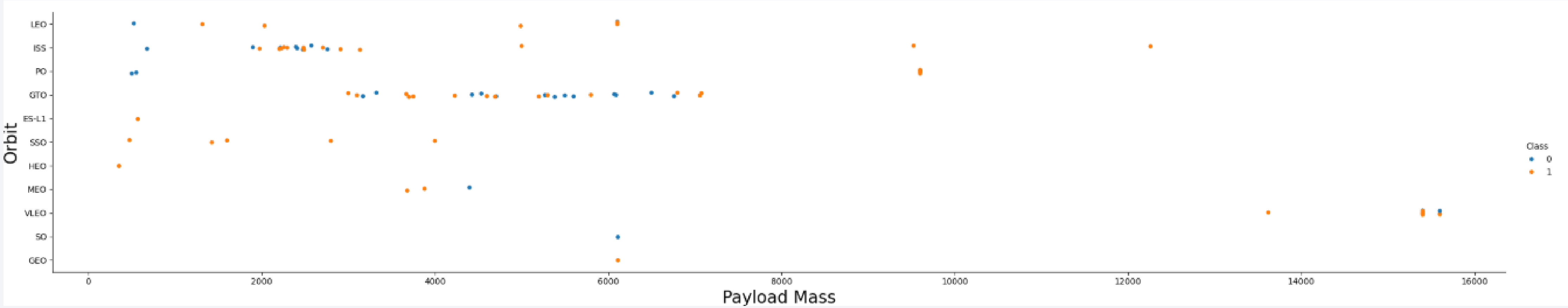
- The plot compares flight number and orbit type against landing outcome.
- More later-flight missions are successful across several orbit categories.
- This supports including Flight Number and Orbit as model features.



External reference: https://github.com/Neuvox512/Data-Science-Capstone-Project

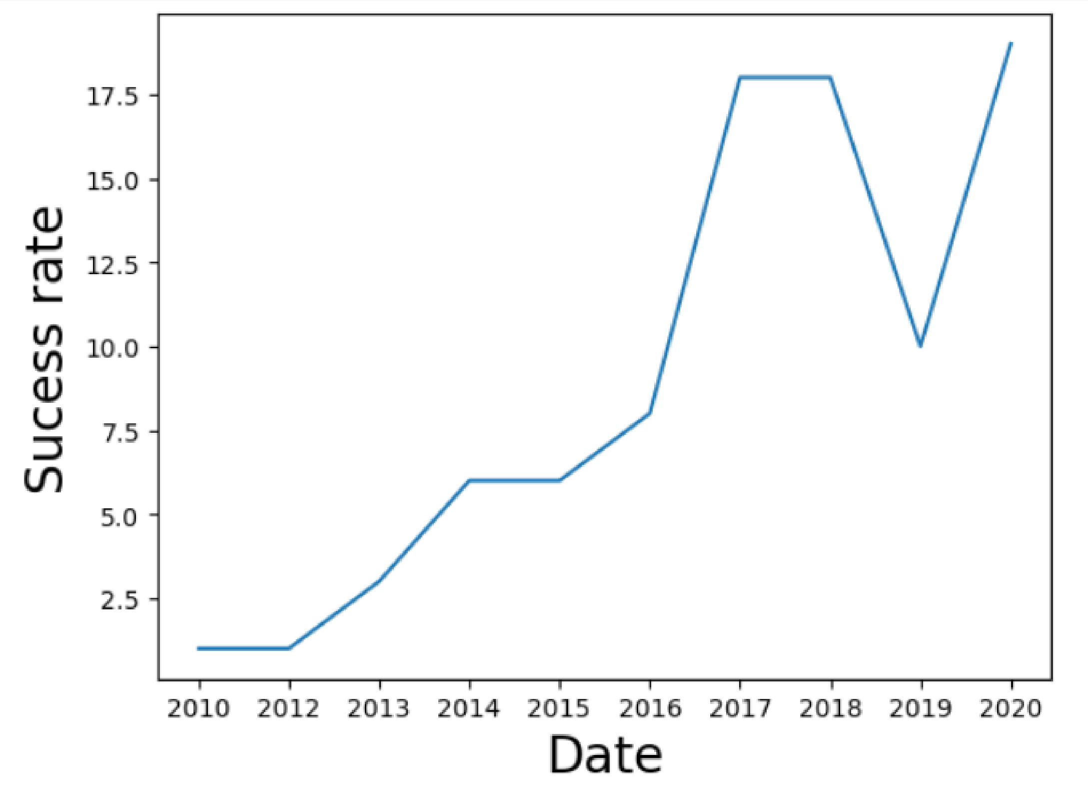
Payload Mass vs. Orbit Type

- Payload mass and orbit type together describe mission difficulty.
- GTO covers a broad middle-to-high payload range.
- Several successful high-payload launches are visible in VLEO.



External reference: https://github.com/Neuvox512/Data-Science-Capstone-Project

Launch Success Yearly Trend



- This slide now uses the notebook output directly.
- The trend rises strongly after 2013, with much higher activity in 2017–2020.
- Note: the uploaded notebook plot reflects yearly launch counts, not a cleaned yearly success-rate chart.

External reference: https://github.com/Neuvox512/Data-Science-Capstone-Project | Visual source: edadataviz.ipynb

All Launch Site Names

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

- Unique names of Launch sites were extracted from database SPACEXTABLE
- SQL query using DISTINCT was implemented for that purpose

Launch Site Names Begin with 'CCA'

- 5 records where launch sites begin with the string 'CCA' were extracted from database SPACEXTABLE
- SQL query using like 'CCA%' LIMIT 5 was implemented for that purpose

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- SQL query using SUM() method was implemented for that purpose
- SQL query also contained condition where customer was chosen as ‘NASA (CRS)’
- SQL result: 45596 kg

Average Payload Mass by F9 v1.1

- SQL query using AVG() method was implemented for that purpose
- SQL query also contained condition where Booster Version was chosen as ‘F9 v1.1’
- SQL result: 2928,4 kg

First Successful Ground Landing Date

- SQL query using MIN() method for date column was implemented for that purpose
- SQL query also contained condition where Landing Outcome was chosen as ‘Success (ground pad)’
- Result date: 2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

- SQL query was used for extraction Booster Version from table
- SQL query also contained 2 conditions:
    - Landing Outcome was chosen as ‘Success (drone ship)’
    - Payload mass was chosen in range between 4000 and 6000 kg

Total Number of Successful and Failure Mission Outcomes

- SQL query was used for extraction Mission Outcome and their count using method COUNT() from table
- GROUP BY method was further implemented in SQL query to get number of Missions to related outcome

Mission_Outcome	count(Mission_Outcome)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

- SQL query was used for extraction Booster Version from table
- Subquery with aggregate function was further implemented in SQL query
- Aggregate function included MAX() method to extract maximum payload mass

2015 Launch Records

- SQL query was used for extraction Booster Version, Launch Site, Year and Month from table
- SQL query also contained 2 conditions:
  - Landing Outcome was chosen as ‘Success (drone ship)’
  - 2015 year was selected

Month	Year	Booster_Version	Launch_Site
01	2015	F9 v1.1 B1012	CCAFS LC-40
04	2015	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Landing_Outcome	count(Landing_Outcome)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

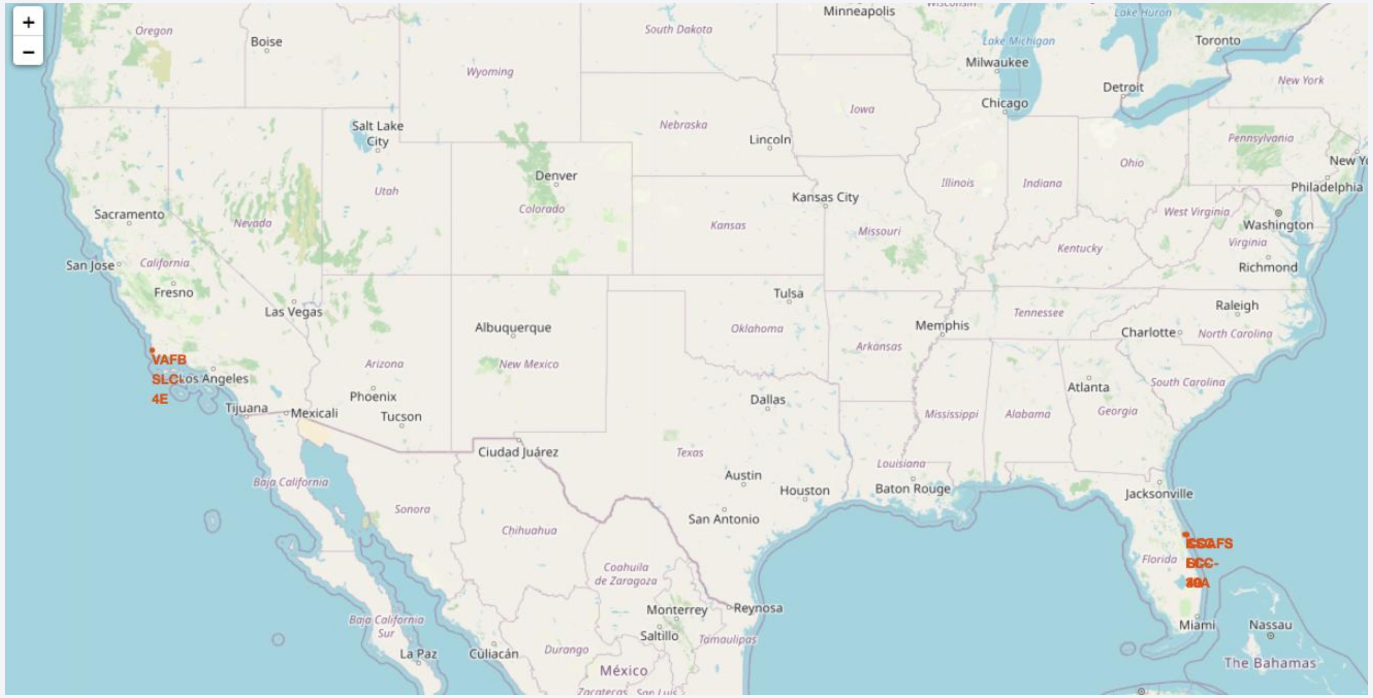
- SQL query was used for extraction Mission Outcome and their count using method COUNT() from table
- GROUP BY method was further implemented in SQL query to get number of Missions to related outcome
- SQL query also included condition where date was selected between 2010-06-04 and 2017-03-20
- The query was further presented by number of successful landings in descending order

Section 3

Launch Sites Proximities Analysis

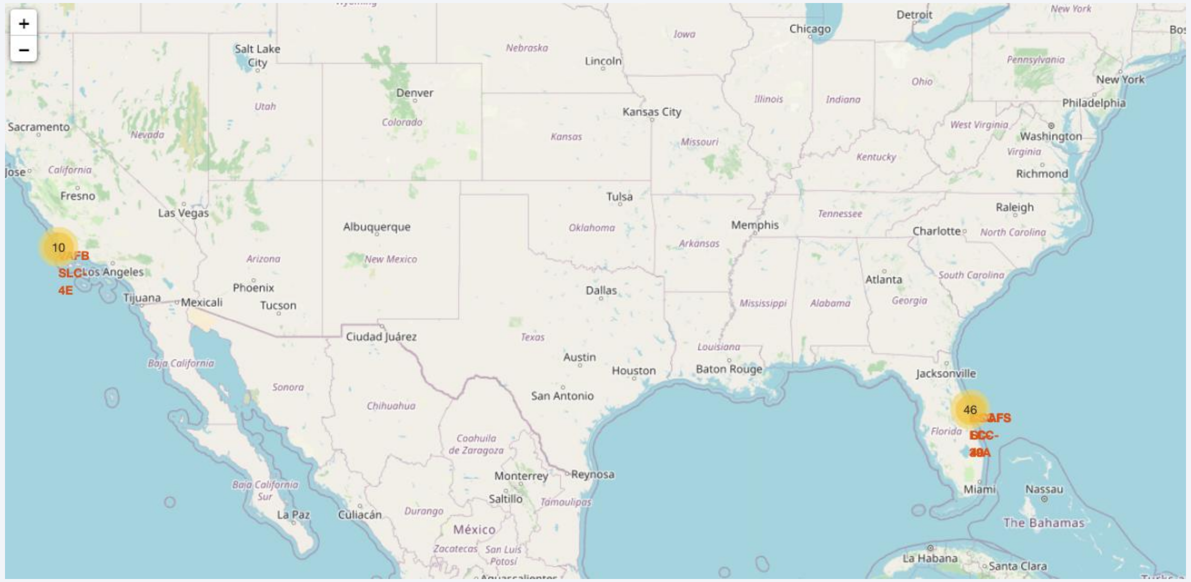
Launch Site Markers

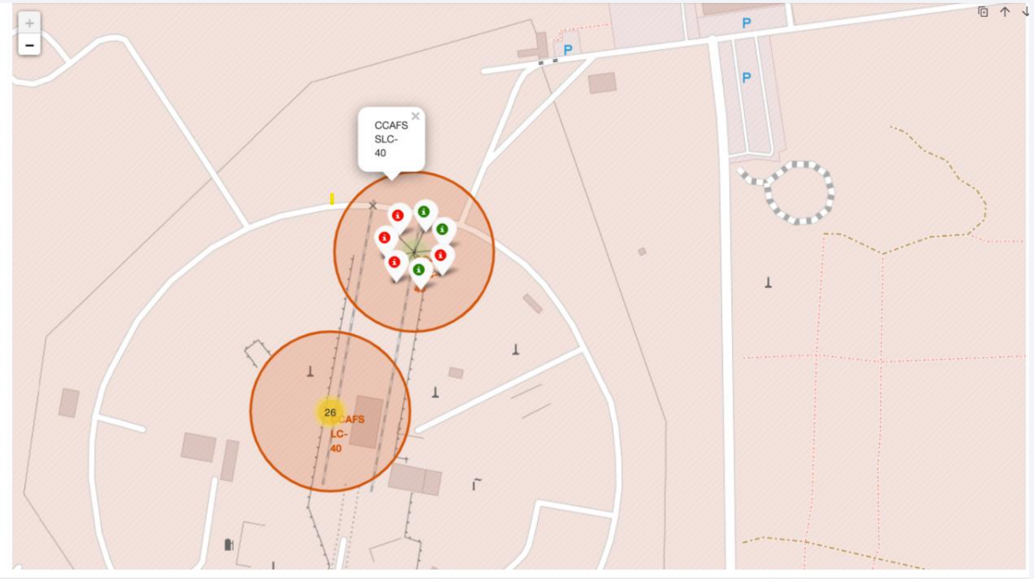
- On global map location markers related to launch site coordinates were placed
- 2 of 3 launch site are settled down very close to each other on east coast



Markers for all launch records

- MarkerCluster object was used to create markers having the same coordinate
- Then we used a green markers for successful launches and red marker for unsuccessful





Launch site proximities

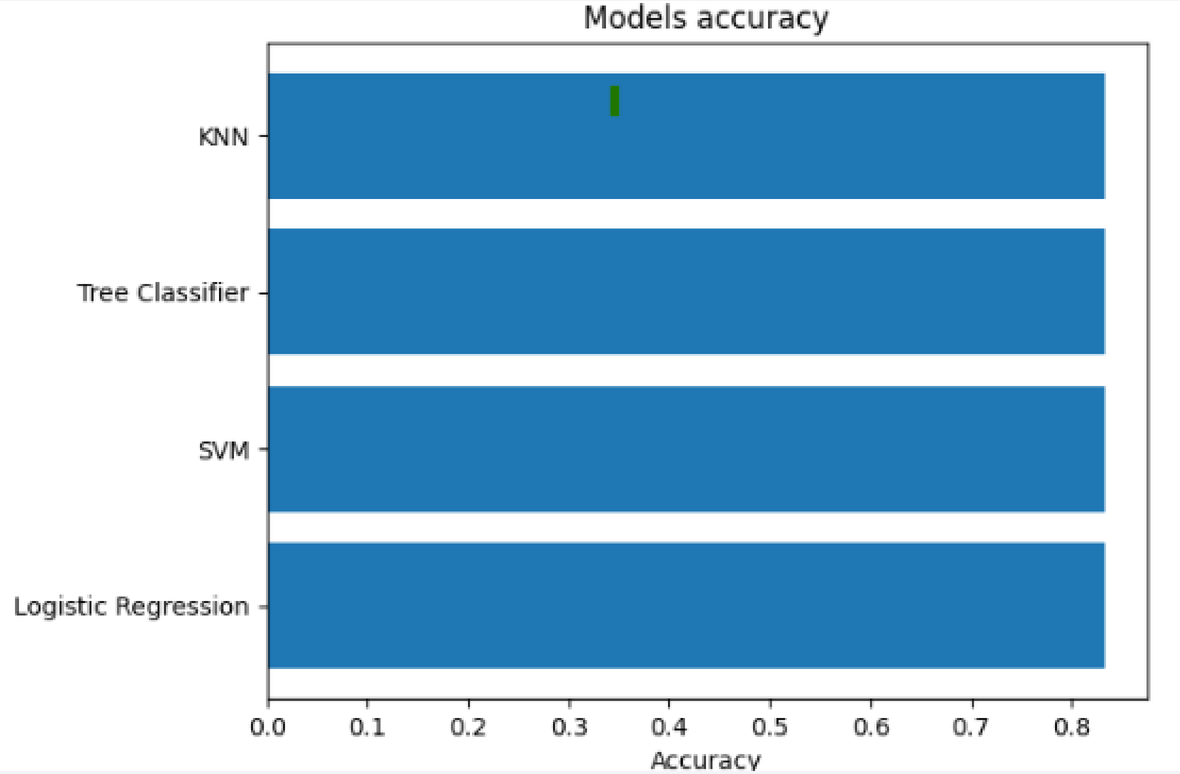
- PolyLine method was
- Point on the closest coastline using MousePosition was marked down and the distance between the coastline point and the launch site was calculated
- Then coastline corresponding Marker was added to the map
- PolyLine method was used to draw the line between launch site and coastline

Section 5

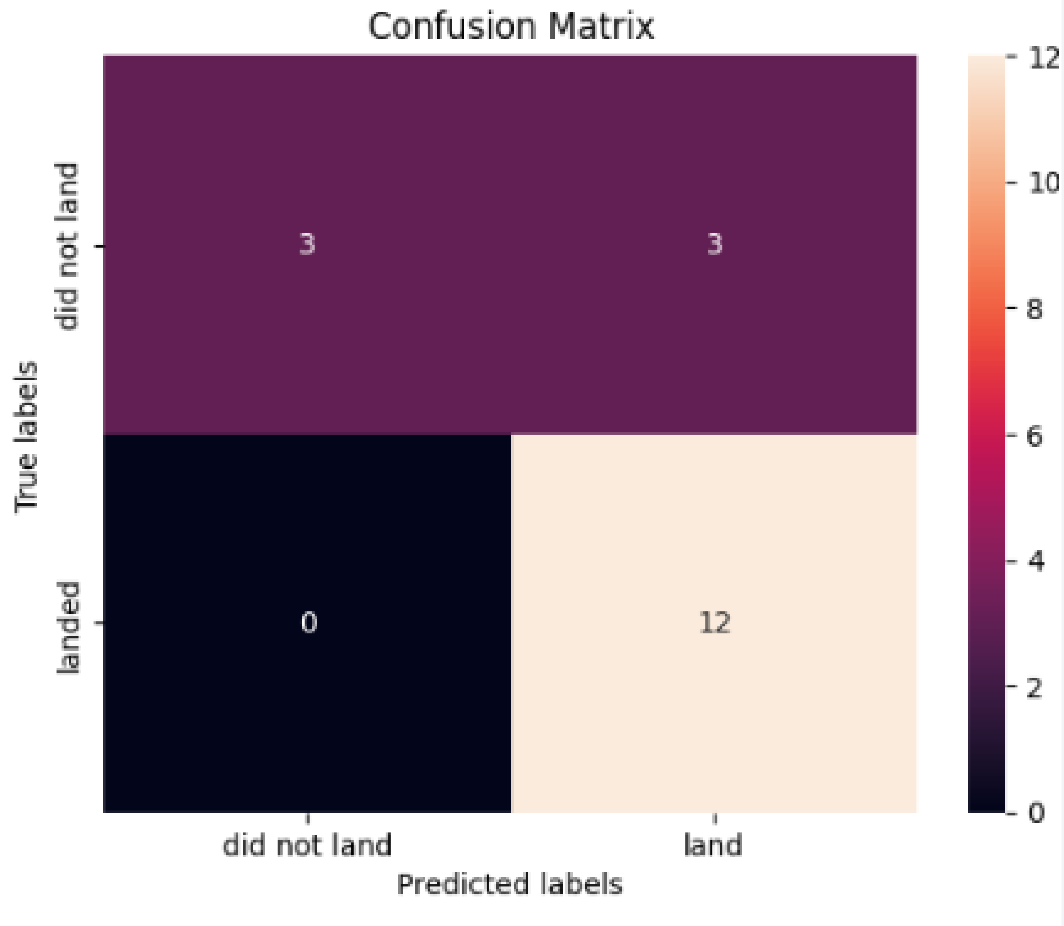
Predictive Analysis (Classification)

Classification Accuracy

- All models performed well in terms of accuracy
- Every model has the same accuracy equal to 83,33 %



Confusion Matrix



- Confusion matrices for all models are the same
- All models have the same False Positive and False Negative rates

Conclusions

1. SpaceX launch cost advantage is tied to first-stage reuse; predicting landing success is therefore commercially important.

2. EDA shows that launch site, orbit type, payload mass, flight number, and year are meaningful predictors of landing outcome.

3. SQL and Folium analysis confirm that launch records are concentrated in a small number of coastal sites with clear operational patterns.

4. Classification models reached comparable accuracy (83.33%); the final recommendation is to use the simplest high-performing model and validate it on new launches.

Thank you!

Appendix: Project Assets

- REST API extraction notebook/code
- Wikipedia scraping notebook/code
- Data wrangling and EDA notebooks
- SQL queries and outputs
- Folium map screenshots
- Predictive analysis notebook and model comparison

Repository: https://github.com/Neuvox512/Data-Science-Capstone-Project